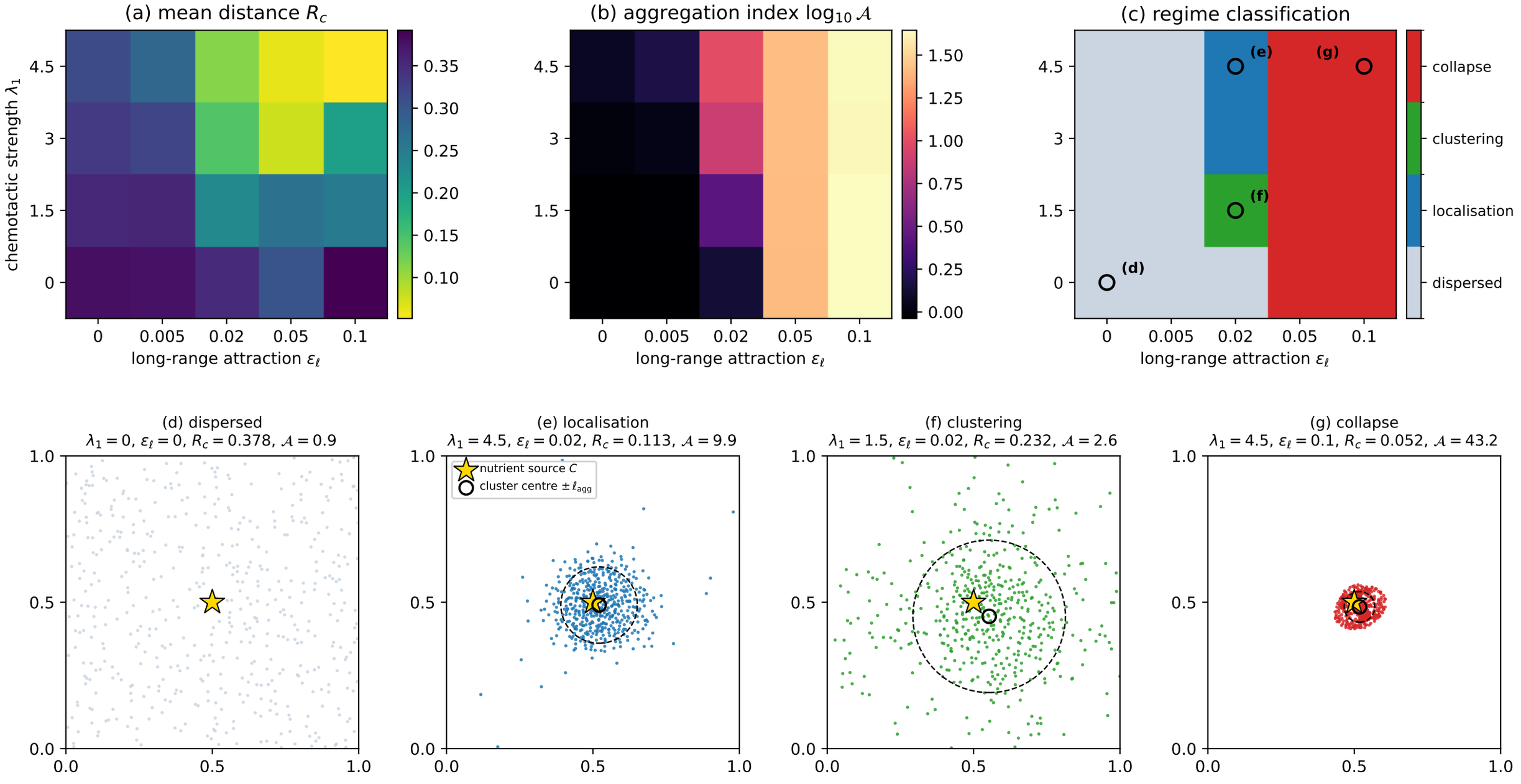

(a) mean distance R_c
(b) aggregation index $\log_{10}\mathcal{A}$
(c) regime classification
chemotactic strength λ_1
long-range attraction ε_ℓ
0
0.005
0.02
0.05
0.1
4.5
3
1.5
0
0.35
0.30
0.25
0.20
0.15
0.10
1.50
1.25
1.00
0.75
0.50
0.25
0.00
(d)
(e)
(f)
(g)
collapse
clustering
localisation
dispersed
(d) dispersed
$\lambda_1 = 0$, $\varepsilon_\ell = 0$, $R_c = 0.378$, $\mathcal{A} = 0.9$
(e) localisation
$\lambda_1 = 4.5$, $\varepsilon_\ell = 0.02$, $R_c = 0.113$, $\mathcal{A} = 9.9$
nutrient source C
cluster centre $\pm \ell_{agg}$
(f) clustering
$\lambda_1 = 1.5$, $\varepsilon_\ell = 0.02$, $R_c = 0.232$, $\mathcal{A} = 2.6$
(g) collapse
$\lambda_1 = 4.5$, $\varepsilon_\ell = 0.1$, $R_c = 0.052$, $\mathcal{A} = 43.2$
0.0
0.5
1.0